

Predicting the Next Category and Region of a Visit: A Check-in-Level Multi-Task Study on Mobility Data

Vitor H. O. Silva, Germano B. dos Santos, Fabrício A. Silva
Laboratório de Inteligência em Sistemas Pervasivos e Distribuídos (NESPED-LAB)
Universidade Federal de Viçosa, Florestal, MG, Brazil
E-mail: {vitor.h.oliveira, germano.santos, fabricio.asilva}@ufv.br

Abstract—Location-based social networks record where people go and what they do, one check-in at a time. If we can anticipate the next visit, mobile and urban services can cache content or reserve capacity ahead of demand. Two coarse questions about the next visit are usually enough: its category (the type of place) and its region (which part of the city). These are normally handled by separate models. We therefore test whether one model can learn both tasks, and what sharing a single model costs in per-task accuracy. We first build a check-in-level representation that describes each visit in its own context, instead of giving every place one fixed vector. Across six datasets (five U.S. states and one non-U.S. city, Istanbul), this improves next-category prediction over a standard place embedding at every dataset, by 0.2 to 6.3 points of macro-averaged F1, with every fold favoring it, and the advantage tracks the geometry of the per-visit vectors rather than the number of training vectors per place. We then train one multi-task model for both tasks and read both answers from a single saved model. On next region it outperforms the dedicated model at the two datasets with the most regions, by about one point, and matches it (statistically, within two points) at the other four. On next category it outperforms the dedicated model at Florida and stays within a fifth of a point of it at the other five. Across both tasks and all six datasets, no difference from a dedicated model exceeds two points, and the model does better where the region task is hardest.

Index Terms—mobility data, next-category prediction, next-region prediction, multi-task learning, check-in-level representation, location-based social networks

I. INTRODUCTION

Location-based social networks, such as Gowalla [1], let people share the places that they visit as check-ins, which together record how people move through a city. Individual mobility is highly predictable in principle [2], so a service that anticipates the next move can prepare ahead of time, caching content where a user is heading [3] or provisioning capacity at the destination before demand arrives. Handover predictions already let cellular services adapt in advance at the network level [4]; at the city scale, check-in mobility analysis is likewise a design input for mobility-aware services [5].

Predicting the next place exactly, a single point of interest (POI), is hard and often more than a service needs. Two coarser questions are usually enough: the next category, the type of place such as food or shopping, and the next region, the part of the city that a user is heading to (a neighborhood-scale unit such as the U.S. census tract). The first captures

intent, the second captures geography; we study whether one model should learn both at once.

Sharing one representation across tasks has a cost: in multi-task learning (MTL), one model does several jobs at once by sharing most of its parts, so the shared parameters can converge to a compromise optimal for neither task, helping one while hurting the other [6]. Our earlier work reported no consistent multi-task advantage for the paired category tasks and attributed it, in part, to this effect [7]; the useful question is where sharing helps, what it costs, and how to share so the gains hold and the cost stays small.

We propose two enhancements. First, we build a check-in-level representation: instead of one fixed vector per place, each check-in gets its own vector that holds its context (the time, nearby places, and recent visits). This adds a fourth level, the check-in, beneath the place, region, and city levels of hierarchical graph infomax [8], [9] (Fig. 1). Second, we train one model that predicts the next category and the next region in a single forward pass, with a shared trunk (a cross-attention stack where the two tasks exchange semantic context) and a private spatial path for the region task (Fig. 2). We evaluate on two settings chosen to differ: five U.S. states of different sizes (Gowalla) and one non-U.S. city (Istanbul).¹

Our contributions are as follows.

- A check-in-level representation that extends hierarchical graph infomax from the place to the individual visit, improving next-category prediction over a standard place embedding at every one of the six datasets, by +0.2 to +6.3 points of macro-F1 (a balanced score that counts every category equally), with every fold favoring it at every dataset (Table III). The advantage tracks the geometry of the per-visit vectors, which separate categories at the visit level and not at the place level. A single fixed vector per place cannot separate places that serve several purposes, and per-visit context recovers that distinction.
- A single model for joint next-category and next-region prediction, sharing semantic context across tasks while keeping a private spatial path for the region task; to our

¹Code (model, representation, baselines, statistical tests): <https://github.com/VitorHugoOli/PoiMtlNet/tree/mobiwac>. Both data sources are public: the category-annotated Gowalla dump (https://figshare.com/articles/dataset/gowalla_data/22126586) and Massive-STEPS [10].

knowledge, the first work to treat fine-grained region as an end target of equal standing (Section II-B). In one forward pass, and reading both answers from a single saved model, it outperforms the dedicated region model at the two datasets with the most regions (by about a point) and stays statistically non-inferior within a two-point margin (TOST, a test of equivalence) at the other four. On category it outperforms the dedicated model at one dataset and stays within a fifth of a point of it at the rest, even though each dedicated model is tuned per dataset (Table II).

- An empirical account, across five U.S. states and one non-U.S. city, of where joint training pays. The two datasets where the joint model outperforms the dedicated region model are the two with the largest region vocabularies (Texas and California), which is where the region task is hardest; at the four smaller vocabularies the two approaches are equivalent within the margin (Fig. 3). The gain does not order monotonically with region count inside that pair, so we report the grouping rather than a law. All joint and dedicated results average four random initializations (Section VI-B).

The rest of the paper presents related work (Section II), the problem (Section III), the method and setup (Sections IV and V), results (Section VI), and discussion and conclusions (Sections VII and VIII).

II. BACKGROUND AND RELATED WORK

A. From place embeddings to per-visit embeddings

A place embedding transforms a POI into a vector so that similar or nearby places sit close together in the vector space. Deep Graph Infomax (DGI) [9], a general self-supervised method for graphs, learns such vectors on a place network linked by similarity, time, and distance, training them to tell the real network from a copy with shuffled node features. Hierarchical Graph Infomax (HGI) [8] adds geographic structure, organizing the network as place, then region, then city. Both produce one vector per place, so two visits to the same coffee shop look identical to the model. Contextual check-in representations instead give each visit its own vector: CTLE [11], the closest example, learns a vector per visit by masking and reconstructing parts of a user’s check-in sequence. Our representation, Check2HGI, also works at the check-in level but through a different construction. CTLE is a sequence model, a Transformer that reads the check-in sequence itself; Check2HGI remains a network model, keeping the hierarchical place-to-region-to-city network and the same infomax objective, now extended one level deeper, from individual places down to individual check-ins (Fig. 1). The order of a user’s visits still enters the network, as links between consecutive check-ins (Section IV-A).

The novelty is this specific combination: per-visit context inside a hierarchical graph-infomax representation. We compare against CTLE and a feature-concatenation control to separate the combination from contextualization alone and from feature injection.

B. Predicting the next category and the next region

A large body of work predicts the next place that a user will visit, the exact POI, with recurrent and attention models such as DeepMove [12], STAN [13], and GETNext [14]. In contrast, we target two properties of the next visit, its category and its region, rather than the exact place. We choose them because they are what a mobility-aware service can act on, not to make the task easier; the region alone spans hundreds to several thousand classes. Predicting over a partition of the map is also the standard formulation in the human-mobility literature, with a grid cell as the target [15]; our next-region task substitutes official neighborhood-scale units for grid cells. Our earlier work [7] paired static category classification with next-category prediction and found no consistent multi-task gain; this paper introduces the next-region task and adds the check-in-level representation, on which sharing helps instead of hurting (Section VI-B).

The field increasingly models several granularities at once; in those systems, category and region are auxiliary signals that help a primary next-place task (MCMG [16], HMT-GRN [17]). We study the pair as the object itself. To our knowledge, fine-grained region as an end target of equal standing, rather than an auxiliary coarse grid cell, is underexplored. The nearest exceptions do not study our exact pairing: DRRGNN [18] forecasts a person’s next activity region jointly with a mobility-intention label, but over regions discovered per person rather than a fixed citywide partition; a generative recommender [19] predicts category and region together, but only as auxiliary steps toward its next-place ranking.

C. Multi-task learning for mobility

MCARNN [20] jointly predicts activity and location, and a recent hierarchy-aware model continues the pattern [21]; in both, the location target is the exact place. The closest alternative to our setting is the cascade: predict the category first, then the place given the category [22]. CSLSL [23] is its current form, predicting in a chain (when, then what, then where), with location as the primary output and category as an intermediate step; CatDM [24] likewise uses the predicted category only to filter candidate places. We instead predict category and region in parallel, as end targets of equal standing in one model with a single forward pass, and we drop the next-place target entirely, so neither task is an intermediate step toward a third. Since the cascade is the pattern that the field uses, we test the choice directly (Section VI-B). On optimization, we are conservative by design: joint training with a fixed loss weighting is standard practice [6], not itself our contribution. Reports find that gradient-balancing methods, which re-weight the tasks’ updates during training (PCGrad [25], Nash-MTL [26]), rarely improve on a well-tuned fixed weighting with two tasks [27]. We confirm this at scale, screening nineteen loss and gradient balancers, including the two methods named above, at their default configurations and a single seed on two datasets, Alabama and Florida: none improved on a tuned fixed task weighting across both

tasks and both datasets. Two exceed equal weighting on next-category at Alabama, Nash-MTL by 0.68 points and scale normalization by 0.19. At Florida, Nash-MTL falls back below equal weighting on both tasks, while scale normalization still exceeds equal weighting on next-category but collapses on next-region. The reason is visible in the gradients. Measured during development on the same joint architecture (on an earlier preparation of the data), the cosine similarity between the next-category and next-region updates on the shared trunk averages $+0.001$ across training (four seeds each on four Gowalla states: Alabama, Arizona and Florida, which are three of the five United States datasets reported here, and Georgia, which this study does not otherwise use; the largest per-dataset mean in absolute value is $+0.0032$). A balancer therefore has no conflict to resolve, consistent with the mechanism test in Section VI-B; this is a finding for this pair of tasks, not a general rule. The same quantity, measured on the final model across seven datasets, is positive at every one of them, so the absence of conflict is not an artifact of this smaller set of runs.

III. PROBLEM AND TASKS

Given a user’s time-ordered check-in history, we predict two properties of the next visit. The first is its *category*, one of seven fixed labels: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. Istanbul’s source collection maps its places onto the same seven labels. The second is its *region*, the place’s census tract (for Istanbul, the *mahalle*, a municipal neighborhood). We do not predict the exact next place; both properties are easier to learn and, for most uses, enough. Region, unlike category, is a large label set: we frame next-region as classification over candidate regions, from 520 classes (Istanbul) to 8,501 (California; Table I).

The motivation is practical: the category says what type of place comes next, hence what a user will want; the region says where, hence where to prepare. The mobility literature supports such preparation: city services have long been built on location-based social network traces [28]. A recent analysis of tourist check-in mobility finds that visits concentrate on a few hub places, argues that crowds and demand therefore concentrate, and ties this structure to infrastructure planning and adaptive strategies [5]. That analysis reads past visits; we predict two properties of the next one, and aggregated over users, those predictions point to where demand is heading before it arrives. A census tract is a neighborhood, not a radio cell. We therefore scope our claims to neighborhood-level preparation: demand and load anticipation, caching content ahead of time, and capacity planning. Cell association and handover are radio-level decisions and remain out of scope. We build and evaluate no such service here; Section VII quantifies what these predictions would give such a service.

IV. METHOD

A. The check-in-level representation

We want each visit, not each place, to have its own vector; we call the resulting representation Check2HGI (Section II-A).

Figure 1 shows the full data flow, from check-ins to the two outputs.

We build a graph with four levels: the check-in, its place, the place’s region (a census tract), and the city. Edges connect each level to the one above it and link a user’s consecutive check-ins with a weight that decays as the time gap between the visits grows; same-place visits connect through their shared place node one level up, and nearby places are linked at the place level. Each visit’s category, time of day, and day of week enter as input features of its node, not as edges. Four elapsed-time features join them: the time since the user’s previous visit and since the user’s first visit, both on a logarithmic scale, the gap to the previous visit within the same day, and an indicator for a user’s first visit. Each is measured up to the visit itself, so a node describes the visit and the history preceding it. These give the representation a sense of tempo, distinguishing a visit that follows minutes after the last one from a visit that opens a new outing, which a categorical hour-and-weekday encoding alone cannot express. This keeps a hierarchical graph’s place-to-region-to-city geography and adds the check-in as a bottom level. We train the graph mainly with an infomax objective, under which each vector learns to match its real neighborhood and reject a shuffled one [8], [9]. Two small label-free auxiliary terms are added (weights 0.3 and 0.1): a masked reconstruction of each place’s aggregated category features, and an anchor to a place embedding pre-trained, label-free, on the same data. The training uses no task label: it never sees the next category or the next region. From the trained graph, we extract one 64-dimensional vector per check-in (its region nodes have trained vectors as well), so a model sees a sequence of per-visit vectors rather than repeated per-place ones.

B. The single multi-task model

We then train one model that reads a window of recent per-visit vectors and predicts the next visit’s category and region at once. Figure 2 shows the design. Each task has its own input. The category task reads the window of per-visit vectors (the semantic stream); the region task reads the same window of visits, each visit now represented by the trained vector of its region node from the same graph (the spatial stream). Both pass through private per-task encoders (a small input network per task, with no shared weights) into the shared trunk, a cross-attention stack of two blocks, so each prediction still uses both inputs. In each block, attention lets each stream read the other’s features, while each keeps its own feed-forward weights. The tasks therefore share by exchanging information between per-task streams, not by owning hidden layers in common. The trunk then feeds a category output and a region output; the region output also keeps a private spatial path: a small branch inside the one model (not a second model) that reads the spatial input window, bypassing the shared trunk. The category task does not touch this branch. Together, the region task’s encoder, its output, and this private path form the region pathway.

The training loss is a fixed-weight sum of the two outputs,

$$L = 0.5 L_{\text{cat}} + 0.5 L_{\text{reg}}, \quad (1)$$

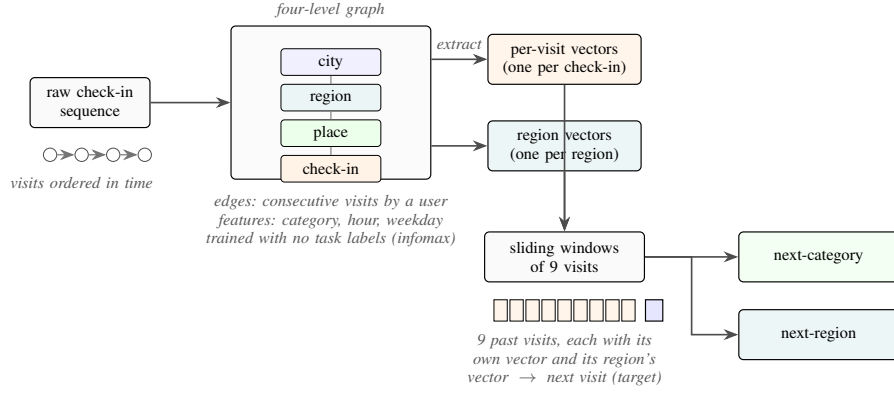


Fig. 1. From a check-in sequence to two predictions: each visit is embedded in a four-level graph (check-in, place, region, city) trained without task labels. The graph yields two sets of vectors, one per visit and one per region; sliding nine-visit windows of each feed a single model that outputs the next category and the next region.

where L_{reg} is a plain unweighted cross-entropy loss and L_{cat} is a cross-entropy loss with logit adjustment: $\tau \log P_{\text{train}}(y)$, with $\tau = 0.5$, is added to the category logits during training only, which shifts the decision boundary toward the balanced posterior that macro-F1 rewards. Inference uses the unadjusted logits. The weights and τ were selected once on validation during development and held fixed across all six datasets. Class weighting, tested on both outputs, lowered both region accuracy and category macro-F1, and logit adjustment replaces it on the category output. This is standard fixed-weight joint training [6], kept simple by design so that any improvement over the dedicated single-task models comes from the shared representation, not from an adaptive weighting scheme. The dedicated category model receives the same adjustment, so the comparison is not affected by it.

The joint model includes the shared trunk and both task outputs, so it is larger than either dedicated model, and a forward pass costs more compute than running the two small dedicated models: the joint model has about 4.2 million parameters at Alabama against 1.1 million for the two dedicated models combined (5.2 against 2.0 at California). What the single model provides is operational rather than arithmetic: one artifact to train, version, and deploy, and one forward pass whose one set of inputs produces both answers at once.

V. EXPERIMENTAL SETUP

This section describes the datasets, data splits, evaluation metrics, statistical tests, and comparison models used in the experiments.

A. Data

Table I summarizes the six datasets. Five datasets come from Gowalla, a location-based social network, and cover the U.S. states of Alabama, Arizona, Florida, Texas, and California [1]. The sixth dataset contains check-ins from Istanbul in the Massive-STEPS collection [10]. We use Istanbul to examine the findings outside the United States. The U.S. datasets range from about 114,000 check-ins and 1,109 regions in Alabama to about 3.2 million check-ins and 8,501 regions in California. All six datasets use the seven place categories defined in

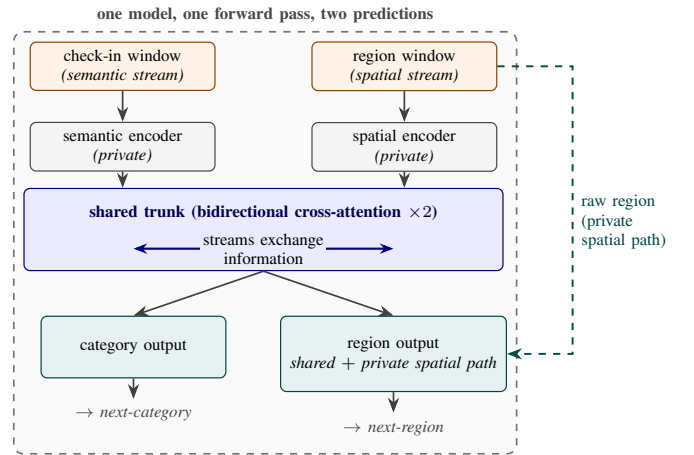


Fig. 2. The single multi-task model. A check-in (semantic) window and a region (spatial) window pass through private encoders into the shared trunk, the only place where the two tasks interact. The category output uses the trunk's output alone; the region output combines it with a private spatial path. One model, one forward pass, two predictions.

Section III. The source labels for Istanbul are mapped to these categories. A region is a census tract in the U.S. datasets and a mahalle in Istanbul.

B. Windows, splitting, and the integrity of the representation

Windows. For each user with at least ten visits, we order the visits by time. We then form overlapping sliding windows of nine visits and use the next visit as the target. A new window starts at each visit, which provides more examples for both tasks. Near the end of a user's history, several start positions can produce shorter, padded windows with the same final visit as the target. We remove these duplicates and keep only the full-length window that ends at this target. The prediction horizon is usually short, but some time gaps are much longer. The median time from the last visit in a window to its target ranges from 0.4 hours in Florida to 5.5 hours in Istanbul. Across the datasets, 5 to 27 percent of the targets occur more than 3 days later.

Splitting. We use stratified five-fold cross-validation and split the data by user. All windows from one user remain in the

same fold. Therefore, the overlapping windows cannot cross the training and test sets, and the visits of a test user never appear in training. The held-out fold provides the validation data, and we do not reserve a third split. We stratify the split by the next-category label because the seven classes are imbalanced. Food is the majority class in every dataset, ranging from about 25 percent of visits in Florida to 34 percent in Alabama (Table I).

Integrity of the representation. We train the representation once on the whole dataset and provide it to every dedicated and joint model. We assess possible information transfer from test visits in three ways. First, the training objective does not use the next-category or next-region targets. It learns by distinguishing observed graph neighborhoods from shuffled ones. Second, we rebuild the representation for each fold using only the training users in that fold. At Alabama, Arizona, and Florida, the resulting differences are at most one third of a point for either task (region, -0.33 to $+0.01$; category, 0.00 to $+0.29$), which is within fold noise. This comparison covers the 67 to 87 percent of visits whose places also appear in training. It does not cover visits to places that are unseen in training. Third, we build the region-transition prior separately for each fold and use only its training data. This prior records how often one region follows another. A version built from the whole dataset inflated region accuracy by 13 to 27 points. Only the HMT-GRN comparison model uses this prior; our joint and dedicated models do not (Section V-D). The other learned representations follow their own training scopes. HGI is pre-trained once on the whole dataset, as our representation is, while CTLE is trained for each fold using training users only.

C. Metrics and statistical tests

For the category task, we report macro-averaged F1 (macro-F1). It is the mean of the F1 scores for the seven categories, so each category has equal weight. Its reference point is the majority-class floor, obtained by always predicting the most common category. For the region task, we report accuracy at ten (Acc@10). This measure is the fraction of test visits for which the true region is among the model’s ten highest-scoring predictions. If the true region does not occur in the training data for that fold, the visit counts as an error. The reference point for this task is the dedicated model.

A claimed gain and a claimed match require different tests. A non-significant difference does not provide evidence of a match. A written analysis plan, fixed during development and before any result was read, assigned a superiority test to next-category prediction and a non-inferiority test to next-region prediction. These assignments apply to each task across the datasets. The plan did not define a superiority test for next-region prediction. Therefore, the two next-region gains in Section VI-B are secondary results outside the plan.

The plan registered a paired Wilcoxon signed-rank test. With four seeds, its exact one-sided p -value cannot be less than 0.0625. We therefore report a paired t -test on the four per-seed means and its 90% confidence interval for the paired

difference. We also report the registered test. The code release includes both tests and documents this departure from the plan.

A *seed* is one complete repetition of the five-fold experiment. It determines the random initialization and the user partition, so each seed produces a different division of the users. Within one seed, the compared models use the same partition. We repeat the experiment across both folds and seeds. Each dataset uses four seeds with five folds per seed, which gives $4 \times 5 = 20$ fitted models per configuration. The statistical tests compare the four per-seed means ($n=4$). We apply a Holm correction [29] across the six next-category comparisons and, separately, across the six next-region comparisons.

The non-inferiority claim states that the joint model is no worse than the dedicated model by more than a two-point margin. We test this claim with the two one-sided tests (TOST) procedure [30]. The analysis plan also fixed the two-point margin in advance. A mobility-aware service acts on which region will be busy, rather than on a single position in the ranking (Section III). A two-point change in Acc@10 is below the level at which this service would behave differently. Across the four user partitions, the standard deviation of the paired difference ranges from 0.02 to 0.16 points. At Istanbul, Arizona, and Florida, the intervals also support a margin as small as one point; at Alabama the interval reaches just past one point, so there the two-point margin is the operative one (Section VI-B).

D. Baselines

We use the comparison models for two purposes: to compare each task with established methods and to evaluate the representation.

Task comparisons. For next-category prediction, we use POI-RGNN [31], which we re-implement from its published architecture and hyperparameters. We also use a Markov model that predicts the category that most often follows the recent categories. We select the best Markov order for each dataset. For region prediction, we compute a Markov-1 floor over region transitions. It uses the same sliding windows and fold splits as our models.

The primary next-region comparison is HMT-GRN [17], an end-to-end recurrent model trained as one unit from the raw check-ins. We evaluate it on the same data, folds, and random initializations. We keep its shared multi-task structure and add a region-transition prior that is built from the training data in each fold. We remove the graph components and hierarchical beam search because they support exact next-place prediction, which we do not study. The resulting model predicts region as one of its original targets, but it is not a reproduction of the complete published system.

We also run STAN [13]. We re-implement its published architecture and train it from the raw check-ins with its own representations and sequence construction under one fixed configuration. We adapt its output to rank candidate regions, which are spatial objects compatible with its candidate-distance matching. We do not adapt STAN to category prediction because distance matching does not apply to categories.

We report ReHDM [32] as a reference under its published protocol.

Representation controls. Two controls test whether the category gain comes from our representation design rather than from contextualization in general or from additional features. CTLE [11] is the closest prior contextual check-in representation. We train it end to end at Florida. At Alabama, Arizona, and Istanbul, we keep its weights fixed and do not fine-tune them. The second control adds raw features from each visit to a standard place embedding and uses the same prediction model.

A final control uses the standard place-level HGI representation [8] to measure what the per-visit representation adds. All other choices, including the nine-visit window, were fixed during development. The released code provides the full training configuration for every model (footnote 1).

VI. RESULTS

A. The check-in-level representation improves the category task

The check-in-level representation outperforms a standard place embedding (HGI [8]) on next-category macro-F1 at every dataset (Table III). The comparison is controlled: target, single-task model, training configuration, windowing, and folds are identical; only the input representation changes (a vector per visit versus a vector per place), so the difference isolates the representation. Both arms also share the epoch budget, the precision, and the logit adjustment, so the only free variable is the input. The gap ranges from +0.23 (Florida) to +6.29 (Istanbul). It is modest in size and unusually consistent in sign: all five folds favor the check-in-level representation at every one of the six datasets, and the difference is significant under a paired test everywhere except Florida ($p = 0.07$). The advantage is a property of the per-visit vectors themselves rather than of the larger number of training vectors per place: the geometry below separates categories at the visit level and not at the place level, which is the distinction the task turns on.

The geometry of the vectors shows why the representation helps this task in particular: the per-visit vectors’ silhouette score by category (how tight and well separated the seven labeled groups are, on a -1 to 1 scale) is about 0.55 against about 0.00 for the place embedding, and their nearest-neighbor category purity (the share of nearest neighbors with the vector’s own category) is about 0.98 against about 0.79 (both averaged over Alabama, Arizona, and Florida). The same geometry does not separate regions, so the benefit is category-only (the joint model’s spatial stream instead reads the region-level vectors of the same graph, Section IV-B). CTLE [11], the closest prior contextual embedding, also gives each visit its own vector, but it pretrains on place identifiers and timestamps alone, so the category vocabulary never enters its training, whereas our graph reads each visit’s category and time of day as input features. At Florida, we fine-tune CTLE together with the task model, using its authors’ defaults and the same 64 dimensions and windowing as ours. It reaches 33.45 macro-F1

at its best epoch and 29.69 at its final one (an epoch is one pass over the training data), below both the place embedding under the same best-epoch rule and the check-in-level representation. With frozen weights, fed to the same single-task model, CTLE repeats the ordering at Alabama, Arizona, and Istanbul. A feature-concatenation control joins the place embedding with the same raw per-visit features that our graph reads (the category one-hot plus hour and day-of-week encodings) and feeds the result to the same single-task model. It is a separate study: a different variant of our graph, its own harness, and embeddings built on different hardware, so its values are not on the scale of Table III and we do not place them beside it. Within that study, concatenation raises the place embedding by $+2.0$, $+1.7$, and $+0.8$ macro-F1 at Alabama, Arizona, and Florida, margins the size of the category gaps we report or larger. On the category axis, therefore, this control does not separate the hierarchical representation from the per-visit features it reads, and we make no such separation claim. On the region task at Florida it moves the place embedding by -0.23 Acc@10, so the injected features do not substitute for the graph on that axis, and the same graph supplies the region-level vectors that the joint model’s spatial stream consumes. We report the category advantage as a measured difference rather than as a decomposition of its causes.

B. One model, two tasks

One model serves both tasks without giving up either. On next region it outperforms the dedicated model at Texas and California, the two datasets with the most regions, and is statistically non-inferior within a two-point margin at the other four. On next category it outperforms the dedicated model at Florida and is within a fifth of a point of it everywhere else. For scale: always predicting the most common category reaches between 5.7 and 7.3 macro-F1 depending on the dataset (the majority-class floor, lowest at Florida, whose category mix is the most even), and a random region top-ten guess is right at most about two percent of the time (Section V-C).

Table II reports the single joint model against the dedicated single-task ceilings (the level a joint model is usually expected, at best, to match). Batch size and learning rate were searched for the dedicated category model at every dataset and for the joint model at the smaller datasets; Texas and California carry the joint configuration transferred from those searches rather than one validated on them, and the dedicated region models use one fixed configuration throughout. Throughout, every reported model is one saved artifact per fold, read at its validation-selected epoch: each dedicated model at its task’s best epoch, and the joint model at the epoch selected by its joint validation score (the geometric mean of the two task metrics), with both tasks read from that one model. Reading each joint task at its own best epoch instead would raise every joint result, by at most 0.17 (category) and 0.90 (region) points, but no single saved model attains both of those numbers at once, which is why the served checkpoint is what we report.

Figure 3 plots the signed differences ordered by region count. On category, the joint model outperforms the dedicated

TABLE I

DATASET STATISTICS, ORDERED BY REGION COUNT: FIVE GOWALLA STATES AND ISTANBUL (MASSIVE-STEPS). ALL SHARE THE SAME SEVEN ROOT CATEGORIES; THE REGION UNIT IS A CENSUS TRACT (A MAHALLE FOR ISTANBUL). WINDOWS: SLIDING NINE-VISIT INPUTS PLUS THE NEXT-VISIT TARGET; LENGTHS: PER-USER CHECK-IN COUNTS; DENSITY: $100 \times \text{CHECK-INS} / (\text{USERS} \times \text{POIS})$; MAJORITY: SHARE OF NEXT-VISIT LABELS IN THE MOST COMMON CATEGORY (FOOD IN EVERY DATASET).

Dataset	Source	Check-ins	Users	POIs	Regions	Windows	Max len	Avg len	Density (%)	Majority (%)
Istanbul	Massive-STEPS	462,615	23,694	29,816	520	271,666	817	19.5	0.065	33.4
AL	Gowalla	113,846	3,858	11,848	1,109	96,326	3,835	29.5	0.249	34.2
AZ	Gowalla	236,450	7,869	20,666	1,547	200,895	5,589	30.1	0.145	34.0
FL	Gowalla	1,407,034	21,052	76,544	4,703	1,274,418	16,679	66.8	0.087	24.7
TX	Gowalla	4,089,892	38,644	160,938	6,553	3,830,414	42,300	105.8	0.066	31.0
CA	Gowalla	3,171,380	37,090	169,145	8,501	2,925,466	14,855	85.5	0.051	32.7

TABLE II

ONE MODEL, TWO TASKS: THE SINGLE JOINT MODEL AGAINST THE DEDICATED SINGLE-TASK MODELS AND THE EXTERNAL BASELINES, ORDERED BY REGION COUNT. CATEGORY IS MACRO-F1; REGION IS ACC@10. BOLD AND $\uparrow$ MARK AN IMPROVEMENT OVER THE DEDICATED MODEL WITH PAIRED STATISTICAL SUPPORT (HOLM-CORRECTED WITHIN EACH TASK; SECTION VI-B). IN THE REGION COLUMNS $\approx$ MARKS A NON-INFERIOR MATCH (TOST, ± 2 PP). CATEGORY CELLS THAT ARE NOT MARKED ARE REPORTED AS MEASURED: THE EQUIVALENCE MARGIN IS PRE-REGISTERED FOR THE REGION AXIS ONLY, SO A CATEGORY DIFFERENCE THAT NO SUPERIORITY TEST RESOLVES IS LEFT UNLABELLED RATHER THAN CALLED A MATCH.

Dataset	Regions	Next-category (macro-F1)				Next-region (Acc@10)			
		Markov-K	POI-RGNN	Dedicated	Joint (ours)	HMT-GRN	ReHDM	STAN	Joint (ours)
Istanbul	520	24.55	30.12	35.34 ± 0.01	35.42 ± 0.06	60.4	69.33	61.86	75.16 ± 0.01
AL	1,109	20.50	23.80	30.77 ± 0.07	30.59 ± 0.07	57.05	65.38	60.72	70.12 ± 0.10
AZ	1,547	23.92	27.64	34.57 ± 0.04	34.57 ± 0.06	43.70	53.00	49.86	59.48 ± 0.07
FL	4,703	29.74	34.49	37.35 ± 0.02	37.55 $\pm 0.07^{\dagger}$	63.74	64.49	72.99	76.69 ± 0.01
TX	6,553	28.67	33.03	36.33 ± 0.01	36.19 ± 0.04	53.85	48.81 ‡	61.67 †	66.15 $\pm 0.07^{\dagger}$
CA	8,501	27.58	31.78	35.63 ± 0.01	35.63 ± 0.02	49.61	50.26 ‡	58.52 †	64.54 $\pm 0.04^{\dagger}$

HMT-GRN (region-native, on our folds, scored on visits whose region appears in training, $>99\%$ of test visits in every dataset and fold) is the primary external comparison; STAN (our re-implementation) and ReHDM (its own protocol) are references. Markov-K: strongest order per dataset. † STAN partial folds: TX 4/5, CA 2/5 (seed 0). ‡ ReHDM at TX and CA: a single seed. Joint and dedicated entries: four seeds $\times$ five folds; $\pm$: sd across seeds. Batch size and learning rate were searched for the dedicated category model at every dataset and for the joint model at the smaller datasets; TX and CA carry the joint configuration transferred from those searches, and the dedicated region models use one fixed configuration throughout.

model at Florida (+0.19 macro-F1) and lands within a fifth of a point of it at the other five datasets, above at Istanbul and below at Alabama, Arizona, Texas, and California. On region (Acc@10), the joint model outperforms the dedicated model at Texas (+1.21) and California (+1.06), the two datasets with the most regions, and stays a non-inferior match (TOST, ± 2 pp) at Istanbul, Alabama, Arizona, and Florida. The two datasets where sharing pays are the two largest region vocabularies, which is consistent with the reading that a shared representation helps most where the region task is hardest; the ordering within that pair does not follow region count, and region count and corpus size co-vary here, so we read the grouping rather than a monotone law.

The statistical picture, seed by seed. On region, Texas and California are the two datasets where all four seeds favor the joint model and the gain survives a Holm correction across the six datasets (paired t on the four per-seed means, Holm-corrected $p = 0.00013$ at Texas and $p = 0.0000063$ at California); their 90% confidence intervals are +1.13 to +1.29 at Texas and +1.03 to +1.08 at California. The remaining four are tested for equivalence (TOST, ± 2 pp) and pass with intervals well inside the margin: Istanbul (-0.08 ; -0.16 to -0.00), Alabama (-0.87 ; -1.00 to -0.75), Arizona (-0.44 ; -0.62 to -0.25), and Florida (-0.16 ; -0.19 to -0.13). All four intervals lie below zero, and three of the four deficits survive a Holm-corrected test in that direction (Alabama, Arizona, and Florida; Istanbul does not); every one of them is far inside two points, so we report them as matches within the margin, not as gains. On category, Florida is the one

dataset where all four seeds favor the joint model and the gain clears the Holm correction (corrected $p = 0.011$; +0.14 to +0.25). The other five category differences are not resolved in either direction by a superiority test. We leave them at that: the equivalence margin of Section V-C is registered for the region axis, and we do not transfer it to category to convert an unresolved difference into a demonstrated match.

A reader used to multi-task learning [6] expects the harder task, here next-region with its hundreds to several thousand classes, to teach the easier one; a control shows otherwise. We freeze the region pathway at the start of training so it can neither learn nor teach the category task. This arm also belongs to a separate study, on a different variant of the graph and its own training configuration, so its absolute scores are not comparable to Table II and we do not report them. What it establishes is directional and about the architecture: within that study, freezing the region pathway leaves the category result unaffected at Alabama, Arizona, and Florida, so the category behavior of this design does not arise from the region task teaching the category one. The control fixes region training rather than any part of the category pathway, so it does not identify which part of the joint architecture produces the region result, and we do not name the shared trunk as its source. Direct ablations of the trunk, at the datasets where they were run, move both tasks by less than two tenths of a point in either direction, which is below what a single-seed five-fold comparison resolves. The decomposition is therefore open, and the trunk is reported as one component of an architecture whose region result is measured, not as the component that

produces it. What the result does establish is operational: both answers come from one model and one forward pass.

The joint model is also above every external baseline reported, on both tasks, across all six datasets (Table II): on region, the primary region-native comparison HMT-GRN [17], STAN [13] trained from the raw check-ins, and a ReHDM reference [32]; on category, POI-RGNN [31] and the Markov-K floor. These externals run on their own embeddings, so this comparison also includes the representation advantage of Section VI-A; the controlled comparison for the joint model remains the Dedicated column of Table II, which uses the same representation, windows, and folds. The first-order Markov region floor of Section V-D, computed under our windows and folds, reaches 51 to 72 Acc@10 across the datasets; the joint model exceeds it by 4.1 to 10.0 points on all six datasets.

We prefer coupling the tasks in parallel rather than in a chain: neither target is subordinate to the other, so a chain must pick one to predict first and pass its errors to the other. The choice still had to be checked, because CSLSL reports its chain outperforming a shared-trunk parallel variant on its own benchmarks [23]. Rewired as a cascade (Section V-D), our model reaches the same combined score, the geometric mean of the two task metrics, as its parallel form: at Alabama, Arizona, Florida, and Istanbul, with both forms trained under the same configuration and initialization, the difference is about zero and neither task moves by more than a quarter of a point. We read this as a defense of the parallel design, not a claim that we outperform the cascade: on this representation the coupling topology makes no measurable difference, and the simpler parallel structure loses nothing. Two qualifications bound this reading: the cascade runs under the configuration tuned for the parallel model, and its form was fixed in advance, so tuning the cascade itself remains future work.

C. External validity (Istanbul)

The joint-model results hold for a non-U.S. city under the same representation, sliding-window protocol, and training setup as the U.S. states. At Istanbul, the joint model is within a tenth of a point of the dedicated model on both tasks: +0.08 macro-F1 on category (dedicated 35.34, joint 35.42) and −0.08 Acc@10 on region (dedicated 75.16, joint 75.08), the latter non-inferior within the two-point margin. The comparable quantity is the difference against the dedicated model, not the absolute Acc@10, since region counts differ across datasets (520 mahalle here). The external baselines show the same ordering on both tasks (Table II). The pattern seen in the U.S. states, one model holding both tasks within the two-point margin of two dedicated models, repeats on a different continent and a different region unit.

VII. DISCUSSION AND LIMITATIONS

A single model serves both tasks in one forward pass, and both answers are read from one saved checkpoint. The region task is where joint training pays: the model outperforms the dedicated region model at Texas and California, the two datasets with the largest region vocabularies, and remains

TABLE III
NEXT-CATEGORY MACRO-F1: THE CHECK-IN-LEVEL REPRESENTATION AGAINST A STANDARD PLACE EMBEDDING (HGI). SAME SINGLE-TASK MODEL, TRAINING CONFIGURATION, FOLDS, SLIDING WINDOWS, EPOCH BUDGET, AND LOGIT ADJUSTMENT ON BOTH ARMS; ONLY THE INPUT REPRESENTATION DIFFERS (SEED 0, FIVE FOLDS).

Dataset	Check-in level	Place level	Δ
Istanbul	35.35	29.07	+6.29
AL	30.77	29.15	+1.62
AZ	34.51	31.93	+2.58
FL	37.36	37.13	+0.23
TX	36.32	35.33	+0.99
CA	35.62	34.74	+0.88

All five folds favor the check-in-level representation at every dataset. The difference is significant under a paired test at every dataset except FL ($p = 0.07$).

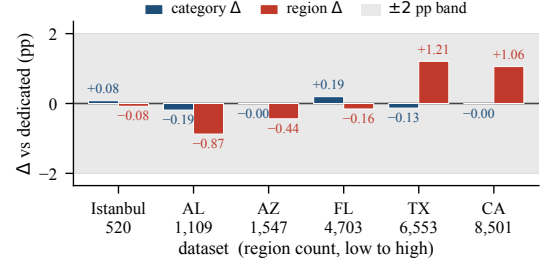


Fig. 3. Signed differences between the joint model and the dedicated single-task models, ordered by region count. Every difference falls inside the ± 2 -point band, which applies to region only. The region difference (Acc@10) is clearly positive at Texas and California, the two largest region vocabularies, and slightly negative at the other four; the category difference (macro-F1) stays within a fifth of a point everywhere.

within the two-point margin at the other four, while preserving a private spatial path for the region output (Table II; Fig. 3). On category it outperforms the dedicated model at Florida and holds within a fifth of a point of it elsewhere.

Two aspects of the design bear on that region result. The shared trunk carries semantic context between the two streams, and the private spatial path keeps the region output free to specialize. Which of them produces the advantage is not decided by the evidence here: direct ablations of the trunk, at the datasets where they were run, move both tasks by less than two tenths of a point in either direction, which is below what a single-seed five-fold comparison resolves. Isolating the trunk at Texas and California, at the full four-seed protocol, is the experiment that would attribute the gain, and it remains open.

The category picture at Texas and California invites a specific hypothesis. Batch size and learning rate were searched for the dedicated category model at every dataset and for the joint model at the smaller datasets; at Texas and California the joint model runs a configuration transferred from those searches rather than one validated on them. Texas carries the second-largest category deficit of the six (−0.13; Alabama’s −0.19 is the largest, and it is a dataset where the joint model was searched), and California’s is under a hundredth of a point. A search conducted for the joint model at Texas and California would test whether the transferred configuration is what holds their category output slightly below the dedicated model. Until it is run this remains a hypothesis rather than an explanation,

and Alabama is a reminder that a searched dataset can show a deficit of the same size.

A service could treat the model’s top-ranked next regions as an anticipatory set: at California, ten regions out of 8,501 contain the true next region 64.54 percent of the time. On four datasets (Alabama, Arizona, Florida, and Istanbul; a single seed over five folds), the ten shortlisted regions lie a median of 3 to 8 kilometers from the shortlist’s centroid, against 17 to 176 kilometers for ten regions drawn at random from the same candidate set (median over 10,000 draws). This remains motivation, not a measured service result.

Five limits qualify these results. First, our representation is trained once over all places; visits to places never seen during training are the single effect that we cannot fully isolate. A planned follow-up trains it on each fold’s training places only and embeds unseen visits. Second, epoch selection consults the fold that the score is then read on (Section V-B), so every absolute score reported here is optimistic. The comparison between the joint model and the dedicated models is affected far less, for two reasons that we can state rather than assume: the selection rule is the same for both models on the same folds, an epoch chosen on validation and read on that fold, with each model selected on its own validation objective (Section VI-B), and, for the category comparison, the dedicated model receives the wider search, a per-dataset sweep over batch size and learning rate at every dataset, against a joint configuration searched at the smaller datasets and transferred to Texas and California, while both sides of the region comparison run a fixed configuration. Where that search applies, the residual favors the dedicated model, which makes the reported category difference conservative. It does not follow that the bias cancels exactly. Third, we do not build or evaluate a mobility-aware service; it is background motivation, and the paper’s claims are the prediction results themselves. Fourth, the joint model carries more parameters than the two dedicated models combined (Section IV-B), and its region pathway in particular is several times the size of the dedicated region model. The region advantage at Texas and California is therefore confounded with capacity: a capacity-matched dedicated region model is the control that would separate the two, and it has not been run. Fifth, each visit node in the representation graph draws on the visits that precede it, so a vector encodes the visit together with its own history; the graph does not pass information from a later visit back to an earlier one, in training or at readout, which is what keeps a node from carrying a feature of the target it is used to predict.

Our representation is a fixed per-visit vector that any downstream model can consume. Moura et al. [5], whose structural analysis of tourist check-ins reads past visits only, name machine learning as a next step; this study moves in that direction.

VIII. CONCLUSION

We tested whether one model can predict both the category and region of a user’s next visit. A check-in-level representation, one vector per visit rather than one per place, predicts

the next category more accurately than the standard place-level representation does, at every dataset and in every fold. On that representation, a single multi-task model reads both answers from one saved checkpoint. On region, it outperforms the dedicated model at Texas and California, the two datasets with the largest region vocabularies, and remains non-inferior (TOST, ± 2 pp) at the other four. On category, it outperforms the dedicated model at Florida and stays within a fifth of a point of it elsewhere. Joint training therefore pays where the region task is hardest, and elsewhere its cost stays inside the margin: the remaining differences are small, several of them resolvable, and none exceeds two points. The joint model also remains above every external baseline in Table II on all six datasets, by at least 3.5 Acc@10 points over the strongest region reference (HMT-GRN, STAN, or ReHDM; the first two on our folds, ReHDM under its own protocol) and by at least 3.0 macro-F1 points over POI-RGNN on category. One model, with a shared semantic context and a private spatial path, anticipates both what a user will do next and where, at one forward pass instead of two, with no difference from a dedicated model exceeding two points on either task.

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